

Autonomous Next-Generation Aerial Refueling System (ANGARS) A-Spec

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1 Introduction

This report intends to;

- Update system specification to accurately reflect the maturity of the current design
- Further categorize requirements into A-Spec-defined binning to more easily trace requirement intent and purpose
- Show the maturity of the system specification design through the resulting quantitative requirements
- Trace newly added functional requirements to system behaviors and Human System Interfacing requirements to their associated user

2 Requirements Metrics

Table 1: Requirements Tracker

	Total	Quantitative	%	Binary	Qualitative
Requirements Analysis Report	154	48	31.1%	63	43
Functional Analysis Report	154	52	33.7%	63	39
Trade Study	159	56	35.2%	63	39
Conceptual Design Report	"	"	"	"	"
Test Plan	"	"	"	"	"
System Specifications	189	145	76.7%	44	0

Through system design maturity the resulting system specification contains 189 requirements, of that 145, or 76.7% are quantitative requirements.

3 A-Spec Requirements

In an attempt to adopt ASpec categorization an additional column was generated (Category) which allocates each requirement to one of the following bins;

- a. General/operational
- b. Overall functional
- c. Overall performance
- d. External interfaces
- e. Subsystem-specific functional/performance

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- f. Internal interfaces
- g. –ilities
- h. Safety

While it's important to add requirement category specificity, it is also important to retain previously defined requirement types (Functional, Interface, Design, etc.) The reason to keep these requirement types is due to the fact that Cameo/MBSE defines and utilizes these requirement types as a tool for model verification. This way, any specific requirement type can be allocated to a given Architectural Element type (interface -> proxyPort, functional -> activityDiagram/Behavior, etc.)

At this level of detail, many requirements are a combination of primary intent (marked as the stereotype of the requirement) and the associated quantitative performance characteristics that supports the intent of the requirement. If the primary intent of a requirement is to specify performance characteristics, then that requirement will be typed as a performance requirement.

3.1 Requirement and Verification Table

3.1.1 External Interface Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
11	ANGARS-11	ANGARS-11 Wingspan Compatibility	The system shall support refueling for autonomous aircraft with wingspans of 5m-15m (T), 5m-18m (O).	Test	Quantitative	☐ false	External interfaces
13	ANGARS-13	ANGARS-13 Multi-Fuel Support	The system shall support ≥ 3 (T), 4 (O) fuel types (eg. Jet A, Jet A-1, Jet B, etc.).	Inspection	Quantitative	☐ false	External interfaces
82	ANGARS-82	ANGARS-82 STANAG 4406 Compliance	The system shall comply with STANAG 4406 for secure messaging to all NATO platforms.	Test	Binary	☐ false	External interfaces
114	ANGARS-114	ANGARS-114 Ground Dashboard Integration	The system shall integrate with ground control station dashboards by providing data outputs formatted in accordance with Aircraft to Ground Command and Data ICD at a rate of at least once every 2 seconds (T), 1 second (O).	Demonstration	Quantitative	☐ false	External interfaces
125	ANGARS-125	ANGARS-125 KC-46 Integration	The system shall integrate with existing KC-46 tankers without structural modifications.	Inspection	Binary	☒ true	External interfaces
126	ANGARS-126	ANGARS-126 Future Tanker Compatibility	The system shall support software updates to maintain compatibility with future tanker models.	Inspection	Binary	☐ false	External interfaces
127	ANGARS-127	ANGARS-127 MIL-STD-1553 Compliance	The system shall comply with MIL-STD-1553 for interfacing with tanker avionics to ensure common interfacing.	Inspection	Binary	☐ false	External interfaces
129	ANGARS-129	ANGARS-129 Tanker Power Compatibility	The system shall interface with tanker power supply outputs of 28VDC nominal.	Inspection	Binary	☐ false	External interfaces
130	ANGARS-130	ANGARS-130 Tanker Fuel Validation	The system shall validate and log tanker fuel quantities against reported values with an accuracy of +/- 2% (T), +/- 1% (O).	Test	Quantitative	☐ false	External interfaces
131	ANGARS-131	ANGARS-131 Tanker Waypoint Sync	The system shall synchronize its navigation data with tanker waypoints within 10m (T), 5m (O).	Inspection	Quantitative	☐ false	External interfaces
132	ANGARS-132	ANGARS-132 Tanker Navigation Interface	The system shall interface with tanker navigation systems in accordance with the Tanker to ANGARS Command and Data ICD.	Inspection	Binary	☐ false	External interfaces
133	ANGARS-133	ANGARS-133 Tanker Health Reporting	The system shall send tanker health data to ground stations every 1 (T), 0.5 (O) seconds.	Analysis	Quantitative	☐ false	External interfaces
134	ANGARS-134	ANGARS-134 Link 16 Compliance	The system shall comply with Link 16 messaging protocols.	Inspection	Binary	☐ false	External interfaces
135	ANGARS-135	ANGARS-135 Tanker Data Bus Auto-Detection	The system shall automatically detect and configure tanker-specific data buses for ≥ 2 (T), 4 (O) distinct tanker data buses.	Inspection	Quantitative	☐ false	External interfaces
136	ANGARS-136	ANGARS-136 Fuel Data Sync	The system shall synchronize fuel data with tanker fuel data logistical systems in accordance with the Aircraft to Ground Command and Data ICD.	Inspection	Binary	☐ false	External interfaces
137	ANGARS-137	ANGARS-137 Tanker Comm Latency	The system shall have a communications interface latency of 5ms (T), 3ms (O) for critical commands.	Test	Quantitative	☐ false	External interfaces
138	ANGARS-138	ANGARS-138 Tanker Integration Size/Weight	The system shall not exceed 750 kg (T), 500 kg (O) or 3 cubic meters (T), 2 cubic meters (O) in size/weight during tanker integration.	Inspection	Quantitative	☐ false	External interfaces
149	ANGARS-149	ANGARS-149 Refueling Request Processing	The system shall receive, authenticate, and process refueling requests from ground control via satellite within 10 seconds (T), 8 seconds (O).	Test	Quantitative	☐ false	External interfaces
187	ASPEC-022	ASPEC-022 A2G and SATCOM BER	The system shall ensure the primary communication link (Air-to-Ground or SATCOM) achieves a Bit Error Rate (BER) of ≤ 10 ⁻⁴ (T), 10 ⁻⁷ (O) under nominal conditions.	Test	Quantitative	☐ false	External interfaces

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3.1.2 General/Operational Requirements

#	Id	Name	Text	Verify Method	KPP	Category	classification
20	ANGARS-18	ANGARS-18 High-Altitude Refueling	The system shall support refueling at altitudes up to 35,000 feet (T), 40,000 feet (O) above mean sea level.	Test	☐ false	General/operational	Quantitative
21	ANGARS-21	ANGARS-21 Adverse Weather Operation	The system shall operate without degradation in operational capability in adverse weather conditions. Adverse weather conditions are defined as; - Rain up to 10mm/hr (T), 15mm/hr (O) - Fog with visibility down to 500m (T), 300m (O) - Crosswinds up to 25 kts (T), 30 kts (O)	Demonstration	☐ false	General/operational	Quantitative
22	ANGARS-26	ANGARS-26 Extreme Temperature Range	The system shall operate in temperature ranges of -35C to +50C (T), -40C to +55C (O).	Demonstration	☐ false	General/operational	Quantitative

3.1.3 -ilities Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
20	ANGARS-20	ANGARS-20 Electronic Warfare Operation	The system shall maintain functionality during electronic warfare conditions up to the jamming levels specified in ANGARS-159.	Analysis	Binary	☐ false	ilities
32	ANGARS-32	ANGARS-32 Anti-Jamming Hopping	The system shall employ frequency-hopping techniques to counter jamming attempts.	Test	Binary	☐ false	ilities
33	ANGARS-33	ANGARS-33 Communication Uptime	The system shall sustain communication uptime of $\geq 99.8\%$ (T), 99.9% (O) between receiving aircraft.	Analysis	Quantitative	☐ false	ilities
35	ANGARS-35	ANGARS-35 Unauthorized Connection Termination	The system shall terminate unauthenticated connections within 1 second (T), 0.5 seconds (O) of authentication failure.	Test	Quantitative	☐ false	ilities
36	ANGARS-36	ANGARS-36 RF Masking	The system shall employ RF signature masking techniques and reduce radar and communication RF emission by ≥ 15 dB (T), ≥ 20 dB (O) during covert operations.	Test	Quantitative	☐ false	ilities
37	ANGARS-37	ANGARS-37 Certificate Validation	The system shall validate digital certificates for all receiving aircraft.	Test	Binary	☐ false	ilities
38	ANGARS-38	ANGARS-38 Message Integrity During Jamming	The system shall achieve $\geq 98\%$ (T), 99% (O) message integrity during jamming.	Test	Quantitative	☐ false	ilities
39	ANGARS-39	ANGARS-39 Zero-Trust Network	The system shall implement a zero-trust network architecture, verifying all access requests regardless of origin, demonstrating compliance with 100% of the core Zero Trust Architecture principles as defined in DoD TBR.	Test	Quantitative	☐ false	ilities
40	ANGARS-40	ANGARS-40 Encryption Recovery	The system shall re-establish encrypted communication within 5 seconds (T), 4 seconds (O) of signal loss.	Test	Quantitative	☐ false	ilities
41	ANGARS-41	ANGARS-41 Tampered Data Erase	The system shall erase sensitive data and notify ground control within 5 seconds (T), 3 seconds (O) if tampering is detected.	Demonstration	Quantitative	☐ false	ilities
42	ANGARS-42	ANGARS-42 Quantum Encryption	The system shall use quantum-resistant encryption algorithms.	Inspection	Binary	☐ false	ilities
43	ANGARS-43	ANGARS-43 AES-256 Encryption	The system shall encrypt all communications using AES-256 encryption protocol.	Inspection	Binary	☒ true	ilities
53	ANGARS-53	ANGARS-53 Collision Expectancy	The system shall have on average, a collision expectancy of no more than once every 1,000,000 mission hours (T), 1,500,000 (O) mission hours.	Analysis	Quantitative	☒ true	ilities
54	ANGARS-54	ANGARS-54 Redundant Pumps	The system shall have at least 1 (T), 2 (O) redundant fuel pumps with automatic switching capability.	Inspection	Quantitative	☐ false	ilities
56	ANGARS-56	ANGARS-56 Triple Redundancy	The system shall employ triple-redundant flight control systems.	Inspection	Quantitative	☒ true	ilities
57	ANGARS-57	ANGARS-57 Backup Power	The system shall have backup power for critical subsystems that are capable of sustaining critical subsystems for ≥ 15 minutes (T), 30 minutes (O).	Test	Quantitative	☐ false	ilities
61	ANGARS-61	ANGARS-61 Redundant IMUs	The system shall include ≥ 2 (T), 3 (O) redundant inertial measurement units (IMUs).	Test	Quantitative	☐ false	ilities
62	ANGARS-62	ANGARS-62 Mission Success Rate	The system shall achieve $\geq 95\%$ (T), 98% (O) mission success rate under nominal conditions.	Analysis	Quantitative	☒ true	ilities
63	ANGARS-63	ANGARS-63 Failure Logging	The system shall log mission failures for post-mission diagnostics with timestamps $\leq 0.5s$ (T), $0.1s$ (O) resolution.	Inspection	Quantitative	☐ false	ilities
64	ANGARS-64	ANGARS-64 Fault Recovery Time	The system shall recover from faults and resume operations within 2 minutes (T), 1 minute (O).	Test	Quantitative	☐ false	ilities
65	ANGARS-65	ANGARS-65 Mission Abort Limit	The system shall limit mission aborts caused by subsystem failures to $\leq 3\%$ (T), 2% (O).	Analysis	Quantitative	☐ false	ilities
66	ANGARS-66	ANGARS-66 Data Archival	The system shall archive mission data for ≥ 30 days (T), 45 days (O) post-operation.	Inspection	Quantitative	☐ false	ilities
69	ANGARS-69	ANGARS-69 False Abort Limit	The system shall limit false-positive mission aborts to no more than 2% (T), 1% (O).	Analysis	Quantitative	☐ false	ilities
71	ANGARS-71	ANGARS-71 Pump Fault Detection	The system shall detect faults in fuel pumps within 5 seconds (T), 2 seconds (O).	Test	Quantitative	☐ false	ilities
72	ANGARS-72	ANGARS-72 Predictive Pump Analysis	The system shall implement algorithms to predict pump failures using vibrational analysis and generate an alert when the calculated probability of failure within the next 50 flight hours reaches a threshold of 50% (T), 75% (O).	Test	Quantitative	☐ false	ilities
74	ANGARS-74	ANGARS-74 Hydraulic Fault Speed	The system shall identify and diagnose hydraulic system faults in ≤ 10 seconds (T), 5 seconds (O).	Analysis	Quantitative	☐ false	ilities
75	ANGARS-75	ANGARS-75 Sensor Isolation	The system shall isolate faulty sensors during diagnostics within 1s (T), 0.5s (O) of fault confirmation by Built-In-Test (BIT).	Demonstration	Quantitative	☐ false	ilities
77	ANGARS-77	ANGARS-77 Log Archive	The system shall archive diagnostics logs for ≥ 1 year (T), 2 years (O).	Demonstration	Quantitative	☐ false	ilities
80	ANGARS-80	ANGARS-80 NATO Cybersecurity	The system shall adhere to NATO cybersecurity standards as defined in ANGARS NATO Cyber Document.	Inspection	Binary	☐ false	ilities
81	ANGARS-81	ANGARS-81 EMI Compliance	The system shall pass electromagnetic interference (EMI) testing.	Inspection	Binary	☐ false	ilities
83	ANGARS-83	ANGARS-83 Penetration Testing	The system shall successfully pass all cybersecurity penetration tests with no exploitable vulnerabilities detected.	Test	Binary	☐ false	ilities
87	ANGARS-87	ANGARS-87 Threat Countermeasures	The system shall employ countermeasures within 2 seconds (T), 1 second (O) of confirmed airborne threat detection.	Test	Quantitative	☐ false	ilities
88	ANGARS-88	ANGARS-88 Breach Containment	The system shall detect and contain cybersecurity breaches by automatically disconnecting impacted systems within 2s (T), 1s (O) of confirmed cybersecurity breach detection.	Demonstration	Quantitative	☐ false	ilities
89	ANGARS-89	ANGARS-89 GPS Spoof Detection	The system shall detect spoofed GPS signals within 6 seconds (T), 2 seconds (O).	Test	Binary	☐ false	ilities

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90	ANGARS-90	 ANGARS-90 Cyberattack System Disabling	The system shall disable non-critical systems during cyberattacks within 2 seconds (T), 1 second (O) to minimize impact from potentially compromised systems.	Test	Binary	☐ false	ilities
91	ANGARS-91	 ANGARS-91 Jamming Mitigation	The system shall mitigate jamming attempts within 30 seconds (T), 15 seconds (O).	Test	Quantitative	☒ true	ilities
92	ANGARS-92	 ANGARS-92 Threat Uptime	The system shall have 99% (T) 99.9% (O) uptime during threat conditions.	Analysis	Quantitative	☐ false	ilities
93	ANGARS-93	 ANGARS-93 Threat Logging	The system shall log all threat responses for post-mission review with time-stamp accuracy of ± 0.1 seconds (T), ± 0.05 seconds (O) for post-mission review.	Inspection	Quantitative	☐ false	ilities
94	ANGARS-94	 ANGARS-94 Countermeasure Updates	The system shall update countermeasure protocols via over-the-air (OTA) updates within 24 hours (T), 12 hours (O) of new threat identification	Analysis	Quantitative	☐ false	ilities
96	ANGARS-96	 ANGARS-96 Annual Autonomy Testing	The system shall meet all requirements of the autonomous mode test protocols each year.	Test	Binary	☐ false	ilities
97	ANGARS-97	 ANGARS-97 Sensor Accuracy Validation	The system shall validate sensor accuracy during testing to $\pm 1\%$ (T), $\pm 0.5\%$ (O) during testing.	Test	Quantitative	☐ false	ilities
98	ANGARS-98	 ANGARS-98 Sandbox Software Testing	The system shall validate all software updates in a sandbox environment before flight implementation.	Test	Binary	☐ false	ilities
99	ANGARS-99	 ANGARS-99 Algorithm Test Coverage	The system shall ensure that all autonomous algorithms are thoroughly tested, achieving 98% (T), 100% (O) test coverage for autonomous algorithms against DO-178C standards and regulations.	Analysis	Quantitative	☐ false	ilities
100	ANGARS-100	 ANGARS-100 Sensor Calibration	The system shall calibrate sensors to 0.5% (T), 0.25% (O) error margin semi-annually.	Demonstration	Quantitative	☐ false	ilities
101	ANGARS-101	 ANGARS-101 Software Checksum	The system shall verify software integrity via checksum check pre-deployment with completion time not exceeding 10 seconds (T), 5 seconds (O).	Inspection	Quantitative	☐ false	ilities
102	ANGARS-102	 ANGARS-102 Hardware Durability Testing	The system shall validate hardware durability over 8,000 (T) operational hours, 10,000 (O) operational hours.	Test	Quantitative	☐ false	ilities
111	ANGARS-111	 ANGARS-111 Operator Action Logging	The system shall log 100% of operator actions. Operator actions include but are not limited to; - button presses - commands - overrides	Inspection	Quantitative	☐ false	ilities
118	ANGARS-118	 ANGARS-118 Component Replacement Time	The system shall support the replacement of critical components to restore operational capability within 4 hours (T), 2 hours (O).	Demonstration	Quantitative	☐ false	ilities
119	ANGARS-119	 ANGARS-119 Modular Subsystems	The system shall employ modular subsystems to enable independent upgrades. Modular subsystems are defined as; - Distinct CIs - Defined, common, interfaces	Inspection	Binary	☐ false	ilities
120	ANGARS-120	 ANGARS-120 Mean Repair Time	The system shall have a mean repair time (MTTR) of 2 hours (T), 1.5 hours (O).	Demonstration	Quantitative	☐ false	ilities
121	ANGARS-121	 ANGARS-121 Obsolescence Management	The system shall implement a minimum of 10-year obsolescence management strategy plan which mitigates the risk of the following items; - high risk components - COTS components	Inspection	Binary	☐ false	ilities
122	ANGARS-122	 ANGARS-122 Lifecycle Cost Estimates	Lifecycle cost estimates for sustainment shall be developed and provided for field maintenance purposes. lifecycle cost estimates include the phases of; - development - production - operations - sustainment - disposal	Analysis	Binary	☐ false	ilities
123	ANGARS-123	 ANGARS-123 Maintenance Training	The system shall include training simulators for maintenance crews that are representative of the system.	Demonstration	Binary	☐ false	ilities
124	ANGARS-124	 ANGARS-124 Repair Documentation	The system shall provide repair documentation covering 100% of the defined intermediate repair procedures	Inspection	Quantitative	☐ false	ilities
128	ANGARS-128	 ANGARS-128 Tanker Integration Time	The system shall maintain integration times with new tanker models within 48 hours (T), 36 hours (O).	Inspection	Quantitative	☐ false	ilities
140	ANGARS-140	 ANGARS-140 Downtime Reporting Accuracy	The system shall report downtime metrics within 2% (T), 1% (O) error.	Inspection	Quantitative	☐ false	ilities
148	ANGARS-148	 ANGARS-148 Predictive Maintenance	The system shall automatically update its maintenance schedule by analyzing performance metrics to determine if maintenance is likely required.	Demonstration	Binary	☐ false	ilities
155	ANGARS-155	 ANGARS-155 System Integration Effort	The system shall require no more than 100 (T), 80 (O) integration man-hours per subsystem interface as measured during system integration testing.	Analysis	Quantitative	☐ false	ilities
156	ANGARS-156	 ANGARS-156 Subsystem TRL	Each subsystem shall achieve a Technology Readiness Level (TRL) of 6 (T), 8 (O) prior to integration.	Analysis	Quantitative	☐ false	ilities
157	ANGARS-157	 ANGARS-157 Subsystem Backup Activation Time	The system shall activate backup subsystems within 2 seconds (T), 1 second (O) of primary subsystem failure.	Analysis	Quantitative	☐ false	ilities
158	ANGARS-158	 ANGARS-158 Backup sensing and navigation	The system shall have one (T), three (O) distinct navigation system redundancy modalities.	Analysis	Quantitative	☐ false	ilities
159	ANGARS-159	 ANGARS-159 Antjam resistance	The system shall maintain functionality through 55 dBm (T), 65 dBm (O) jamming attempts.	Test	Quantitative	☐ false	ilities
160	ANGARS-160	 ANGARS-160 Sensor Fusion Switch Time	The system shall automatically switch between active sensor inputs (radar, LIDAR, and M-code GPS) within 1.5 seconds (T), 1.0 second (O) upon detecting sensor performance degradation.	Test	Quantitative	☐ false	ilities
162	ANGARS-162	 ANGARS-162 Antenna Redundancy Failover Time	The system shall detect a primary antenna degradation and switch to the secondary antenna within 500 ms (T), 300 ms (O) of failure detection.	Test	Quantitative	☐ false	ilities

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163	ANGARS-163	ANGARS-163 Fuel Pump Redundancy Activation Time	The system shall activate a redundant fuel pump and maintain continuous fuel transfer within 3 seconds (T), 2 seconds (O) after detecting a primary pump fault.	Test	Quantitative	☐ false	ilities
164	ANGARS-164	ANGARS-164 Power Subsystem Failover Time	The system shall detect a primary power failure and switch to backup UPS and distribution within 2 seconds (T), 1 second (O).	Test	Quantitative	☐ false	ilities
166	ASPEC-001	ASPEC-001 System MTBCF	The system shall achieve a Mean Time Between Critical Failures (MTBCF) of ≥ 500 (T), 750 (O) flight hours. Critical Failure is defined as mission abort or safety impact	Test	Quantitative	☐ false	ilities
167	ASPEC-002	ASPEC-002 System Operational Availability	The system shall achieve an Operational Availability (Ao) of $\geq 99.0\%$ (T), 99.5% (O) during scheduled mission periods. Ao is calculated as $\text{Uptime} / (\text{Uptime} + \text{Downtime})$.	Analysis	Quantitative	☐ false	ilities
168	ASPEC-003	ASPEC-003 AGNS Subsystem MTBCF	The AGNS subsystem shall demonstrate an MTBCF of ≥ 1000 (T), 1500 (O) flight hours. Critical Failure is defined as mission abort or safety impact	Test	Quantitative	☐ false	ilities
169	ASPEC-004	ASPEC-004 Subsystem MTTR	The Mean Time To Repair (MTTR) for any subsystem shall be ≤ 2.0 (T), 1.5 (O) hours.	Demonstration	Quantitative	☐ false	ilities
170	ASPEC-005	ASPEC-005 BIT Fault Isolation	The ANGARS system Built-In-Test (BIT) shall detect $\geq 98\%$ of critical failures and isolate $\geq 90\%$ (T), 95% (O) of detected critical failures to a single subsystem.	Test	Quantitative	☐ false	ilities
176	ASPEC-011	ASPEC-011 Mission Reliability	The system shall achieve a mission reliability of $\geq 98\%$ (T), 99.5% (O) for a standard 10-hour refueling mission profile.	Analysis	Quantitative	☐ false	ilities
177	ASPEC-012	ASPEC-012 Failure Isolation Time	The system shall isolate detected critical failures impacting safety or mission success to a single subsystem within 5 minutes (T), 2 minutes (O) of BIT completion.	Test	Quantitative	☐ false	ilities
178	ASPEC-013	ASPEC-013 Maintenance Alerts	The system shall track the operational hours/cycles for 100% of the limited-life components and generate maintenance alerts when their predefined thresholds are reached.	Test	Quantitative	☐ false	ilities

3.1.4 Internal Interface Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
59	ANGARS-59	ANGARS-59 Dual Data Buses	The system shall employ dual (T), triple(O)-redundant data buses.	Inspection	Quantitative	☐ false	Internal interfaces
161	ANGARS-161	ANGARS-161 Flight Control Synchronization	The system shall ensure that the triple-redundant flight control modules synchronize command execution within 100 ms (T), 50 ms (O) of each other.	Test	Quantitative	☐ false	Internal interfaces
165	ANGARS-165	ANGARS-165 Processing Data Fusion Latency	The system shall fuse and process sensor data from the AGNS subsystem and update system status within 500 ms (T), 300 ms (O) under nominal operational conditions.	Test	Quantitative	☐ false	Internal interfaces
181	ASPEC-016	ASPEC-016 Data Bus Loading	The system shall ensure loading on critical internal data buses does not exceed 70% (T), 60% (O) of the maximum rated bus capacity during simulated peak operational scenarios.	Test	Quantitative	☐ false	Internal interfaces
182	ASPEC-017	ASPEC-017 Time Sync Accuracy	The system shall maintain time synchronization across critical command, data, and timing subsystems (AGNS, C&C, Processing, Comms) with an accuracy of ± 1 millisecond (ms) (T), ± 0.1 millisecond (ms) (O) relative to actual time.	Test	Quantitative	☐ false	Internal interfaces
183	ASPEC-018	ASPEC-018 End-to-End C&C Latency	The system shall ensure end-to-end data latency for critical fused sensor data from AGNS subsystem output to the C&C subsystem input does not exceed 20 ms (T), 10 ms (O).	Test	Quantitative	☐ false	Internal interfaces
184	ASPEC-019	ASPEC-019 C&C to Fuel Transfer System Latency	The system shall ensure the command latency from C&C subsystem output to Fuel Transfer Boom Actuator response is ≤ 150 ms (T), 100 ms (O).	Test	Quantitative	☐ false	Internal interfaces
185	ASPEC-020	ASPEC-020 Internal Ethernet BER	The system shall ensure the Bit Error Rate (BER) on the primary internal Ethernet data bus does not exceed 10^{-9} (T), 10^{-8} (O) under nominal conditions.	Test	Quantitative	☐ false	Internal interfaces

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3.1.5 Overall Functional Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
1	ANGARS-1	ANGARS-1 Autonomous Refueling	The system shall execute autonomous refueling via onboard sensors without human assistance.	Demonstration	Binary	☐ false	Overall functional
8	ANGARS-8	ANGARS-8 Adaptive Fuel Speed	The system shall adjust refueling speed within a range of +/- 20% (T), +/- 30% (O) from nominal based on receiving aircraft stability.	Analysis	Quantitative	☐ false	Overall functional
10	ANGARS-10	ANGARS-10 Multi-Platform Refueling	The system shall employ refueling procedures for both manned and unmanned aircraft as defined in the system's configuration manual.	Test	Binary	☐ false	Overall functional
12	ANGARS-12	ANGARS-12 Fuel Flow Adaptation	The system shall adjust fuel flow rate the fuel flow rate by +/- 15% (T), +/- 25% (O) from the nominal rate based on receiving aircraft type and properties.	Demonstration	Quantitative	☐ false	Overall functional
17	ANGARS-17	ANGARS-17 Altitude by Aircraft Type	The system shall automatically adjust its altitude based on receiving aircraft type within 60 seconds (T), 45 seconds (O) of receiving aircraft mission information.	Analysis	Quantitative	☐ false	Overall functional
19	ANGARS-19	ANGARS-19 GPS-Denied Navigation	The system shall operate in GPS-denied environments by automatically switching to alternate navigation methods within 2 seconds (T), 1 second (O) of GPS signal loss.	Test	Quantitative	☐ false	Overall functional
22	ANGARS-22	ANGARS-22 Inertial Navigation	The system shall automatically switch to and utilize inertial navigation when GPS is not available (GPS denial, outages) within 0.5s (T), 0.25s (O) of GPS outage.	Demonstration	Quantitative	☐ false	Overall functional
23	ANGARS-23	ANGARS-23 Low-Visibility Sensors	The system shall utilize non-GPS sensors (LIDAR, RADAR, optical) for positioning and docking when GPS accuracy may be degraded.	Test	Binary	☐ false	Overall functional
24	ANGARS-24	ANGARS-24 Mission Task Prioritization	The system shall automatically prioritize mission-critical tasks over non-critical tasks during contested operations such that operational tasking can be executed within $\leq 100\text{ms}$ (T), $\leq 50\text{ms}$ (O) latency.	Analysis	Quantitative	☐ false	Overall functional
28	ANGARS-28	ANGARS-28 Threat Zone Avoidance	The system shall automatically calculate and execute flight paths around active and known threat zones in accordance with received mission tasking.	Analysis	Binary	☐ false	Overall functional
29	ANGARS-29	ANGARS-29 Satellite Relay	The system shall reroute mission data through relay satellites communication path if direct ground links are jammed within 5s (T), 10s (O) of jamming detection.	Test	Quantitative	☐ false	Overall functional
30	ANGARS-30	ANGARS-30 Pre-Comm Authentication	The system shall authenticate all receiving aircraft before data exchange within 4 seconds (T), 3 seconds (O) of initial contact.	Test	Quantitative	☐ false	Overall functional
44	ANGARS-44	ANGARS-44 Obstacle Detection Range	The system shall detect obstacles within 500m (T), 750m (O) with 99% accuracy.	Test	Quantitative	☒ true	Overall functional
49	ANGARS-49	ANGARS-49 Foreign Object Detection	The system shall detect birds/foreign objects $\geq 10\text{cm}$ (T), 7cm (O) wide.	Test	Quantitative	☐ false	Overall functional
51	ANGARS-51	ANGARS-51 Predictive Obstacles Algorithms	The system shall employ predictive algorithms for obstacle trajectories and avoidance capable of estimating detected obstacle trajectories $\geq 5\text{s}$ (T), 7s into the future.	Analysis	Quantitative	☐ false	Overall functional
67	ANGARS-67	ANGARS-67 Post-Mission Reports	The system shall generate post-mission performance reports including key performance data upon mission completion and send this to Ground Control within 20s (T), 15s (O) from time of mission completion. Key performance data includes and is defined as; - Fuel transferred - Duration - Alerts - Anomalies	Inspection	Quantitative	☐ false	Overall functional
68	ANGARS-68	ANGARS-68 Status Notifications	The system shall notify operators of mission success/failure within 1 minute (T), 30 seconds (O).	Demonstration	Quantitative	☐ false	Overall functional
70	ANGARS-70	ANGARS-70 Fuel Compatibility Check	The system shall validate fuel compatibility between receiving aircraft within 3 seconds (T), 2 seconds (O) before transfer.	Test	Quantitative	☐ false	Overall functional
86	ANGARS-86	ANGARS-86 Jamming Detection	The system shall detect jamming signals within 10 seconds (T), 8 seconds (O) and transmit alerts to ground control.	Demonstration	Quantitative	☐ false	Overall functional
95	ANGARS-95	ANGARS-95 Threat Alert Priority	The system shall prioritize threat alerts over non-critical communications during contested operations with transmission delay not exceeding 0.5 seconds (T), 0.3 seconds (O) during contested operations. Threat alerts are defined as; - EW Detection - Mission warning - Adversarial aircraft (Identify Friend Foe (IFF) non-compliant) - Imminent collision	Analysis	Quantitative	☐ false	Overall functional
141	ANGARS-141	ANGARS-141 Fuel Schedule Optimization	The system shall employ an optimized fuel transferring schedule based on receiving aircraft priority. Parameters to include in optimization algorithm include; - aircraft priority - fuel state - time constraints - efficiency factors	Analysis	Binary	☐ false	Overall functional
145	ANGARS-145	ANGARS-145 Fuel Quality Check	The system shall verify fuel quality before transfer.	Inspection	Binary	☐ false	Overall functional
147	ANGARS-147	ANGARS-147 Low-Fuel Priority	The system shall prioritize refueling for aircraft with a customizable pre-defined critical threshold before other aircraft in queue.	Analysis	Binary	☐ false	Overall functional
150	ANGARS-150	ANGARS-150 Refueling Priority Logic	The system shall prioritize refueling requests based on aircraft fuel urgency, mission criticality, and operational efficiency algorithms.	Analysis	Binary	☐ false	Overall functional
152	ANGARS-152	ANGARS-152 Dynamic Schedule Updates	The system shall update schedules dynamically in response to new requests or mission disruptions (e.g., threats, weather) or aircraft aborts.	Test	Binary	☐ false	Overall functional

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3.1.6 Overall Performance Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
2	ANGARS-2	ANGARS-2 Refueling Time	The system shall complete refueling within 10 minutes (T), 8 minutes (O) of handshake (alignment) with receiving aircraft.	Test	Quantitative	☒ true	Overall performance
4	ANGARS-4	ANGARS-4 Aircraft ID Verification	The system shall verify and validate receiving aircraft identity within 5 seconds (T), 3 seconds (O) before beginning fuel transfer.	Demonstration	Quantitative	☐ false	Overall performance
5	ANGARS-5	ANGARS-5 Turbulence Stability	The system shall maintain a stable connection with the receiving aircraft during turbulence with disconnection rate not exceeding 2% (T), 1% (O) during turbulent refueling.	Test	Quantitative	☐ false	Overall performance
14	ANGARS-14	ANGARS-14 Fuel Capacity Check	The system shall verify receiving aircraft fuel capacity before transfer within 3 seconds (T), 2 seconds (O) of initial connection.	Demonstration	Quantitative	☐ false	Overall performance
25	ANGARS-25	ANGARS-25 Terrain Navigation	The system shall employ terrain-based navigation with position accuracy of $\pm 20m$ (T), $\pm 15m$ (O) when operating in GPS-denied environments.	Demonstration	Quantitative	☐ false	Overall performance
27	ANGARS-27	ANGARS-27 Contested Positioning Accuracy	The system shall maintain positional accuracy of $\leq 5m$ (T), $4m$ (O) in contested environments.	Test	Quantitative	☒ true	Overall performance
31	ANGARS-31	ANGARS-31 Low-Latency Commands	The system shall maintain $\leq 15ms$ (T), $10ms$ (O) latency in command transmission.	Test	Quantitative	☒ true	Overall performance
34	ANGARS-34	ANGARS-34 Data Loss Limit	The system shall have data packet loss of $\leq 0.2\%$ (T), 0.1% (O).	Test	Quantitative	☐ false	Overall performance
47	ANGARS-47	ANGARS-47 Collision Update Frequency	The system shall update collision risk assessments every 0.5 seconds.	Test	Quantitative	☐ false	Overall performance
50	ANGARS-50	ANGARS-50 False Alert Limit	The system shall limit false collision alerts to no more than two (T) one (O) per mission.	Analysis	Quantitative	☐ false	Overall performance
139	ANGARS-139	ANGARS-139 Fuel Transfer Rate	The system shall transfer fuel at 1,200 gallons/minute (T), 3000 gallons/minute(O).	Test	Quantitative	☐ false	Overall performance
142	ANGARS-142	ANGARS-142 Fuel Waste Limit	The system shall limit fuel waste to no more than 1.0% (T), 0.5% (O) of the total fuel consumed in each mission	Analysis	Quantitative	☐ false	Overall performance
143	ANGARS-143	ANGARS-143 Fuel Transfer Accuracy	The system shall ensure that actual fuel transfer volumes are within 5.0% (T), 3.0% (O) of the planned values.	Test	Quantitative	☐ false	Overall performance
146	ANGARS-146	ANGARS-146 Transfer Completion Reporting	The system shall report fuel transfer completion within 5 seconds (T), 10 seconds (O).	Demonstration	Quantitative	☐ false	Overall performance
151	ANGARS-151	ANGARS-151 Schedule Generation Speed	The system shall generate a refueling schedule and broadcast it to queued aircraft and the ground within 30 seconds (T), 20 seconds (O) of mission assignment.	Demonstration	Quantitative	☐ false	Overall performance
153	ANGARS-153	ANGARS-153 Post-Mission Reporting	The system shall generate and transmit post-mission reports (fuel transferred, anomalies, threats) to ground control within 5 minutes (T), 3 minutes (O) of mission completion.	Demonstration	Quantitative	☐ false	Overall performance
154	ANGARS-154	ANGARS-154 Mission Status Updates	The system shall transmit mission status updates to ground control every 30 seconds.	Demonstration	Quantitative	☐ false	Overall performance

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3.1.7 Safety Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
3	ANGARS-3	ANGARS-3 Proximity Safety Abort	The system shall abort refueling within 0.5 seconds (T), 0.2 seconds (O) if the receiving aircraft proximity exits established safe bounds.	Test	Quantitative	☐ false	Safety
7	ANGARS-7	ANGARS-7 Fuel Leak Detection	The system shall detect fuel leaks ≥ 0.1 gal/min (T), ≥ 0.05 gal/min (O) off nominal expectations and cease fuel transfers within 2s (T), 1s (O) of detection.	Test	Quantitative	☐ false	Safety
45	ANGARS-45	ANGARS-45 Collision Rerouting	Upon detection of a collision risk exceeding the defined threshold, the system shall initiate an avoidance maneuver within 1s (T), 0.5s (O).	Analysis	Quantitative	☐ false	Safety
46	ANGARS-46	ANGARS-46 Aircraft Separation	The system shall maintain ≥ 100 m (T), 150m (O) separation from non-receiving aircraft.	Test	Quantitative	☐ false	Safety
48	ANGARS-48	ANGARS-48 Collision Avoidance Priority	The system's decision logic mechanisms shall prioritize collision avoidance over mission timelines.	Analysis	Binary	☐ false	Safety
52	ANGARS-52	ANGARS-52 Terrain Avoidance	The system shall avoid terrain obstacles located below 500m (T), 750m (O) above ground level (AGL)	Test	Binary	☐ false	Safety
55	ANGARS-55	ANGARS-55 Power Loss Safe Mode	The system shall default to safe mode within 0.5 seconds (T), 0.3 seconds (O) of detecting power loss.	Test	Quantitative	☐ false	Safety
58	ANGARS-58	ANGARS-58 Hydraulic Isolation	The system shall automatically detect and isolate hydraulic failures within 2 seconds (T), 1 second (O) to prevent fault propagation.	Demonstration	Quantitative	☐ false	Safety
60	ANGARS-60	ANGARS-60 Boom Fail-Safe Locks	The system shall engage fail-safe mechanical locks for the refueling boom within 1 second (T), 0.5 seconds (O) of emergency condition detection.	Inspection	Quantitative	☐ false	Safety
78	ANGARS-78	ANGARS-78 FAA Compliance	The system shall comply with FAA airworthiness standards as defined in the ANGARS Certification Plan.	Inspection	Binary	☐ false	Safety
79	ANGARS-79	ANGARS-79 MIL-STD-882E Compliance	The system shall meet or exceed MIL-STD-882E safety protocols, including hazard analysis and risk assessment.	Inspection	Binary	☐ false	Safety
84	ANGARS-84	ANGARS-84 ICAO Annex 6 Compliance	The system shall meet or exceed ICAO Annex 6 safety requirements.	Test	Binary	☐ false	Safety
85	ANGARS-85	ANGARS-85 DO-178C Compliance	The system shall comply with DO-178C for software airworthiness.	Analysis	Binary	☐ false	Safety
105	ANGARS-105	ANGARS-105 Emergency Shutdown	The system shall include emergency shutdown controls allowing for system-wide emergency shutdown within 1s (T), 0.5s (O) of activation.	Test	Quantitative	☐ false	Safety
115	ANGARS-115	ANGARS-115 Critical Control Lockdown	The system shall disable non-essential controls during critical emergencies within 0.5s (T), 0.2s (O) of emergency state confirmation.	Test	Quantitative	☐ false	Safety
171	ASPEC-006	ASPEC-006 Emergency Fuel Cutoff Time	The system shall activate the emergency fuel transfer cutoff sequence within $\leq 0.2s$ (T), 0.1s (O) of receiving a cutoff command or detecting a critical anomaly.	Test	Quantitative	☐ false	Safety
172	ASPEC-007	ASPEC-007 Safety Program Compliance	The system safety program shall be conducted in accordance with MIL-STD-882E, including hazard analysis and risk assessment.	Analysis	Binary	☐ false	Safety
173	ASPEC-008	ASPEC-008 Sensor Conflict Safety Logic	The system shall prioritize maintaining safe separation based on the in the event of conflicting sensor data regarding collision detection.	Test	Binary	☐ false	Safety

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3.1.8 Subsystem-Specific Functional/Performance Requirements

#	Id	Name	Text	Verify Method	classification	KPP	Category
6	ANGARS-6	ANGARS-6 Probe Alignment Accuracy	The system shall align the refueling probe within 0.5m (T), 0.25m (O) of the receiving aircrafts fueling port.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
9	ANGARS-9	ANGARS-9 Probe Retraction	The system shall fully retract the refueling probe within 10s (T), 7s (O) of refueling completing.	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
15	ANGARS-15	ANGARS-15 Altitude Boom Adjustment	The system shall automatically adjust the boom length to compensate for altitude differences in receiving aircraft.	Demonstration	Binary	☐ false	Subsystem-specific functional/performance
16	ANGARS-16	ANGARS-16 Fuel Port Adaptation	The system shall automatically calibrate and adapt to account for variations in fuel port configurations on receiving aircraft within a positional range of +/- 10cm (T), +/- 5cm (O) relative to nominal.	Analysis	Quantitative	☐ false	Subsystem-specific functional/performance
73	ANGARS-73	ANGARS-73 Fuel Temperature Monitoring	The system shall monitor fuel temperature within 2C (T), 1C (O) accuracy.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
76	ANGARS-76	ANGARS-76 Low Pressure Alert	The system shall alert operators of low fuel pressure within 1 second (T), 0.5 seconds (O) of detection.	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
103	ANGARS-103	ANGARS-103 HMI Mission Display	The system shall make available via HMI to operators at minimum at a rate of at least once every 1 second (T), 0.5 seconds (O); - real-time mission schedules - queued aircraft status (ID, priority, fuel state) - active threat alerts - overall system health status	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
104	ANGARS-104	ANGARS-104 Manual Override Speed	The system shall allow manual override via HMI within 2.5 seconds (T), 2 seconds (O).	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
106	ANGARS-106	ANGARS-106 Tactile Alerts	The system shall provide tactile feedback for critical alerts within 0.3 seconds (T), 0.2 seconds (O) of alert generation.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
107	ANGARS-107	ANGARS-107 Voice Command Support	The system shall support voice commands for operator actions with a command recognition accuracy of ≥ 90% (T), ≥ 95% (O) in a standard cockpit noise environment of TBR.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
108	ANGARS-108	ANGARS-108 HMI Refresh Rate	The system shall display graphical data at 30 Hz (T), 60 Hz (O) via HMI.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
109	ANGARS-109	ANGARS-109 Subsystem Health Display	The system shall display subsystem health using red (fault), yellow (degraded), and green (nominal) per MIL-STD-1472G.	Inspection	Binary	☐ false	Subsystem-specific functional/performance
110	ANGARS-110	ANGARS-110 Multilingual HMI	The system shall include the option for multilingual support for operators supporting up to 3 (T), 4 (O) additional languages.	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
112	ANGARS-112	ANGARS-112 Haptic Emergency Alerts	The system shall provide haptic feedback to operators for emergency alerts.	Demonstration	Binary	☐ false	Subsystem-specific functional/performance
113	ANGARS-113	ANGARS-113 HMI Update Speed	The system shall update status displays every 0.5 seconds (T), 0.2 seconds (O).	Test	Quantitative	☐ false	Subsystem-specific functional/performance
116	ANGARS-116	ANGARS-116 Manual Override Instructions	The system shall provide step-by-step instructions for manual override within 10 (T), 5 (O) seconds of operator request.	Analysis	Quantitative	☐ false	Subsystem-specific functional/performance
117	ANGARS-117	ANGARS-117 Refueling Queue Reprioritization	The system shall provide an interface accessible to authorized operators to manually reprioritize refueling queues.	Demonstration	Binary	☐ false	Subsystem-specific functional/performance
144	ANGARS-144	ANGARS-144 Turbulence Minimization	The system shall implement boom control algorithms to reduce boom oscillation displacements by ≥ 30% (T), 50% when compared to uncontrolled measurements.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
174	ASPEC-009	ASPEC-009 HMI Display Contrast Ratio	All text and critical symbology displayed on the Operator HMI panel shall maintain a minimum contrast ratio of 7:1 (T), 10:1 (O) under ambient lighting from 100 to 1000 lux.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
175	ASPEC-010	ASPEC-010 Manual Control Latency	The system shall have no more than ≤ .25s (T), .2s (O) between the time an operator inputs for manual boom control to result in corresponding boom actuator response.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
179	ASPEC-014	ASPEC-014 HMI Point Size	The system shall ensure critical warning and caution alert text displayed on the HMI uses a minimum font size of 18 points (T), 20 points (O).	Demonstration	Quantitative	☐ false	Subsystem-specific functional/performance
180	ASPEC-015	ASPEC-015 HMI Feedback	The system shall ensure its HMI provides clear, unambiguous feedback to the operator indicating the reason for an invalid command or data entry rejection within 1 second (T), 0.5 seconds (O).	Test	Quantitative	☐ false	Subsystem-specific functional/performance
186	ASPEC-021	ASPEC-021 C&C Loading	The system shall ensure the C&C subsystem processor utilization does not exceed 80% (T), 70% (O) during simulated peak operational workload scenarios.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
188	ASPEC-023	ASPEC-023 Voltage Regulation	The system shall ensure the main 28VDC power bus provided by the Power Subsystem maintains voltage regulation within ± 3% (T), 2% (O) of nominal.	Test	Quantitative	☐ false	Subsystem-specific functional/performance
189	ASPEC-024	ASPEC-024 Non-Critical HMI Response	The system shall provide HMI response to operator inputs within 500 ms (T), 250 ms (O) for non-critical functions.	Test	Quantitative	☐ false	Subsystem-specific functional/performance

3.2 Human System Interface (HIS) Requirements and User Allocations Matrix

Legend			
	Allocate		
			ANGARS Physical Context
			Air Force Mission Planner
			Operator in the Loop
	2.1.0 All Requirements [2.1 Mission Requirements]	1	16
	ANGARS-103 HMI Mission Display	1	
	ANGARS-104 Manual Override Speed	1	
	ANGARS-106 Tactile Alerts	1	
	ANGARS-107 Voice Command Support	1	
	ANGARS-108 HMI Refresh Rate	1	
	ANGARS-109 Subsystem Health Display	1	
	ANGARS-110 Multilingual HMI	1	
	ANGARS-112 Haptic Emergency Alerts	1	
	ANGARS-113 HMI Update Speed	1	
	ANGARS-115 Critical Control Lockdown	1	
	ANGARS-116 Manual Override Instructions	1	
	ANGARS-117 Refueling Queue Reprioritization	1	
	ASPEC-009 HMI Display Contrast Ratio	1	
	ASPEC-010 Manual Control Latency	1	
	ASPEC-014 HMI Point Size	1	
	ASPEC-015 HMI Feedback	1	
	ASPEC-024 Non-Critical HMI Response	1	

As the system is primarily an autonomous system, it makes sense that there are so few requirements that would interact with, at some capacity, an external actor. For this reason, we see a limited number, but not too few, linked to an external actor. Most of these requirements are defining how the human/actor physically or strategically interacts with the system.

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3.3 KPPs

#	Name	Derived From	classification	Text	Verify Method	KPP
1	ANGARS-2 Refueling Time	N1 Autonomous Refueling	Quantitative	The system shall complete refueling within 10 minutes (T), 8 minutes (O) of handshake (alignment) with receiving aircraft.	Test	true
2	ANGARS-27 Contested Positioning Accuracy	N3 Contested Environment Operation	Quantitative	The system shall maintain positional accuracy of $\leq 5m$ (T), 4m (O) in contested environments.	Test	true
3	ANGARS-31 Low-Latency Commands	N4 Secure Communication	Quantitative	The system shall maintain $\leq 15ms$ (T), 10ms (O) latency in command transmission.	Test	true
4	ANGARS-43 AES-256 Encryption	N3 Contested Environment Operation N4 Secure Communication N10 Threat Detection & Response	Binary	The system shall encrypt all communications using AES-256 encryption protocol.	Inspection	true
5	ANGARS-44 Obstacle Detection Range	N5 Collision Avoidance	Quantitative	The system shall detect obstacles within 500m (T), 750m (O) with 99% accuracy.	Test	true
6	ANGARS-53 Collision Expectancy	N5 Collision Avoidance	Quantitative	The system shall have on average, a collision expectancy of no more than once every 1,000,000 mission hours (T) 1,500,000 (O) mission hours.	Analysis	true
7	ANGARS-56 Triple Redundancy	N6 Redundant Fail-Safe Systems	Quantitative	The system shall employ triple-redundant flight control systems.	Inspection	true
8	ANGARS-62 Mission Success Rate	N7 Mission Reliability	Quantitative	The system shall achieve $\geq 95\%$ (T), 98% (O) mission success rate under nominal conditions.	Analysis	true
9	ANGARS-91 Jamming Mitigation	N10 Threat Detection & Response	Quantitative	The system shall mitigate jamming attempts within 30 seconds (T), 15 seconds (O).	Test	true
10	ANGARS-125 KC-46 Integration	N14 Integration with Tanker Platforms	Binary	The system shall integrate with existing KC-46 tankers without structural modifications.	Inspection	true

4 Functional Traceability

Legend Satisfy		2.3 Logical Behaviors ANGARS Top Level ANGARS Top Level System Behaviors	
2.1.0 All Requirements (2.1 Mission Requirements)		2.3 Logical Behaviors	
ANGARS-11 Wingspan Compatibility	1 1	F1: Manage Refueling Requests (context: Crypto Module)	1
ANGARS-12 Fuel Flow Adaptation	1 1	F1.1: Receive & Authenticate Request (context: Crypto Module)	1
ANGARS-13 Multi-Fuel Support	1 1	F1.2: Validate Fuel Capability (context: Flight Processing Module)	1
ANGARS-14 Fuel Capacity Check	1 1	F1.3: Prioritize Requests (context: Flight Processing Module)	1
ANGARS-15 Altitude Boom Adjustment	1 1	F1.4: Generate Schedule Dynamically (context: Flight Processing Module)	1
ANGARS-16 Fuel Port Adaptation	1 1	F1.5: Update Schedule Dynamically (context: Flight Processing Module)	1
ANGARS-17 Altitude by Aircraft Type	1 1	F1.6: Transmit Status & Reports	1
ANGARS-18 High-Altitude Refueling	1 1	F2: Control Autonomous Docking	1
ANGARS-19 Pump Fault Detection	1 1	F2.1: Receive Aircraft Telemetry	1
ANGARS-20 Predictive Pump Analysis	1 1	F2.2: Adjust Boom Length & Altitude	1
ANGARS-21 Fuel Temperature Monitoring	1 1	F2.3: Calibrate to Fuel Port (context: Refueling Boom Controller)	1
ANGARS-22 Hydraulic Fault Speed	1 1	F2.4: Align Refueling Probe (context: Refueling Boom Controller)	1
ANGARS-23 Sensor Isolation	1 1	F2.5: Stabilize During Turbulence	1
ANGARS-24 Low Pressure Alert	1 1	F2.6: Retract Probe	1
ANGARS-25 Log Archive	1 1	F2.7: Manage Docking	1
ANGARS-26 HMI Mission Display	1 1	F3: Execute Fuel Transfer	1
ANGARS-27 Manual Override Speed	1 1	F3.1: Validate Fuel Compatibility	1
ANGARS-28 Emergency Shutdown	1 1	F3.2: Confirm Fuel Type	1
ANGARS-29 Tactile Alerts	1 1	F3.3: Initiate Fuel Transfer	1
ANGARS-30 Voice Command Support	1 1	F3.4: Monitor Flow & Stability (context: Fuel Flow Meter & Pressure)	1
ANGARS-31 HMI Refresh Rate	1 1	F3.5: Emergency Shutdown	1
ANGARS-32 Subsystem Health Display	1 1	F3.6: Complete Transfer (context: Fuel Pumps)	1
ANGARS-33 Multilingual HMI	1 1	F3.7: Adjust Flow Rate	1
ANGARS-34 Refueling Queue Reprioritization	1 1	F4: Monitor Safety & Diagnostics	1
ANGARS-35 Tanker Fuel Validation	1 1	F4.1: Monitor Proximity	1
ANGARS-36 Tanker Waypoint Sync	1 1	F4.2: Check Fuel Integrity (context: Fuel Detection Sensor)	1
ANGARS-37 Backup sensing and navigation	1 1	F4.3: Log Failures and Archive Data	1
ASPEC-000 Emergency Fuel Cutoff Time	2 2	F4.4: Manage Fuel Recovery (context: Flight Control Module)	1
ASPEC-001 Sensor Conflict Safety Logic	2 2	F4.5: Monitor Fuel Sensor Position and Control Mode	1
ASPEC-002 Maintenance Alerts	2 2	F4.6: Send Status Notifications	1
		F4.7: Power System	1
		F5: Manage Communications	1
		F5.1: Receive Real-Time Statuses	1
		F5.2: Receive Mission Data	1
		F5.3: Receive Maintenance Operations and Files	1
		F5.4: Send Data	1
		F5.5: Activate Comms Countermeasures (context: Antenna Control)	1
		F5.6: Alert System (context: Haptic Alert Unit)	1
		F5.7: Analyze Environment (context: Active Sensor Suite)	1
		F5.8: Analyze GPS (context: Flight Processing Module)	1
		F5.9: Analyze Radar (context: Active Sensor Suite)	1
		F5.10: Send Threat Status	1
		F6: Operate in Adverse Conditions	1
		F6.1: Validate Digital Certificates (context: Crypto Module)	1
		F6.2: Authenticate Aircraft (context: Crypto Module)	1
		F6.3: Terminate Unauthorized Connections (context: Crypto Module)	1
		F6.4: Zero-Trust & RF Masking	1
		F6.5: Apply Cryptography (context: Crypto Module)	1
		F6.6: Ensure Msg Integrity (context: Crypto Module)	1
		F6.7: Re-establish Encryption (context: Crypto Module)	1
		F6.8: Erase Data	1
		F6.9: Send Threat Mitigation	1
		F7: Enforce Cybersecurity	1
		F7.1: Display Mission Data (context: HMI Panel & Displays)	1
		F7.2: Receive Operator Input	1
		F7.3: Process Manual Override (context: Operator Control Panel)	1
		F7.4: Execute Emergency Controls (context: Flight Control Module)	1
		F7.5: Provide Alerts & Feedback (context: Haptic Alert Unit)	1
		F7.6: Update HMI Displays (context: HMI Panel & Displays)	1
		F7.7: Subsystem Health & Multilingual Interface (context: HMI Panel & Displays)	1
		F7.8: Logging & Dashboard Integration (context: HMI Panel & Displays)	1
		F7.9: Reprioritize Refueling Queue (context: Flight Processing Module)	1
		F8: Manage HMI	1
		F8.1: Distribute Power (context: Power Supply)	1
		F8.2: Receive Maintenance Operations & Flight Processing Power	1

Appendix A – Schedule and EVM Updates

Schedule

Table 2: Capstone Schedule

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
1	Project Concept & Proposal	50	10-Dec	21-Jan	25	10-Dec	1/19/2025
1.1	Intro	5					
1.3	System Need	5					
1.4	System Description	15					
1.5	Background	10					
1.6	Justification	5					
1.7	Draft Cleanup & Submittal	5					
1.8	Final Revisions & Submittal	5					
2	Requirements Analysis Report (RAR) & CONOPS	45	21-Jan	2-Feb		21-Jan	2-Feb
2.1	Requirements Definition & Description	8					
2.2	Requirements Verification Plan	6					
2.3	User/Stakeholder Needs Analysis & Report	7					
2.4	Key Performance Parameters (KPP) Identification & Compilation	6					
2.5	CONOPS - Concept Diagram	6					
2.6	CONOPS - Potential Users/Operators List	4					
2.7	CONOPS - Use Case Scenario Definition	5					
2.8	RAR & CONOPS Cleanup & Submittal	3					
3	Functional Analysis Report (FAR)	37	2-Feb	16-Feb			
3.1	Functional Tree Definition	5					
3.2	Functional Decomp / Tree	5					
3.3	Context Diagram UPDATES	3					
3.4	Function List Development	4					
3.5	Functional Flow Diagrams Creation	6					
3.6	N2 Diagrams Development	5					
3.7	Requirements-Functions Traceability Matrix	4					
3.8	System States & Modes Definition	3					
3.9	FAR Cleanup & Submittal	2					
4	Trade Study	32	16-Feb	23-Feb			
4.1	Trade Study Purpose Definition	3					
4.2	Selection Criteria Development	4					
4.3	Alternatives Analysis	5					
4.4	Criteria Weights Development	4					
4.5	Utility Functions Development	4					
4.6	Raw Criteria Values Collection	4					
4.7	Weighted Utility Calculations	3					
4.8	Sensitivity Analysis	3					
4.9	Trade Study Cleanup & Submittal	2					

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WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
5	Concept Design Report	37	23-Feb	2-Mar			
5.1	Physical Context Diagram Development	5					
5.2	Physical Component Tree Creation	4					
5.3	Subsystem Descriptions	5					
5.4	Physical Block Diagrams Development	6					
5.5	Physical N2 Diagrams Creation	5					
5.6	Data Flow Diagrams	4					
5.7	Interface Definitions & Descriptions	3					
5.8	Functions-Components Mapping	3					
5.9	Concept Design Cleanup & Submittal	2					
6	Test Plan	27	2-Mar	9-Mar			
6.1	Test Objectives Definition	3					
6.2	Requirements-Test Objectives Traceability	4					
6.3	Test Environment Description	3					
6.4	Test Equipment Identification	3					
6.5	Test Subjects & Articles Definition	3					
6.6	Success Criteria Development	3					
6.7	Metrics Definition	3					
6.8	Test Execution Planning	3					
6.9	Test Plan Cleanup & Submittal	2					
7	A-Specification Report	22	9-Mar	16-Mar			
7.1	Requirements Update & Organization	4					
7.2	KPP Updates	3					
7.3	Support Requirements Development	3					
7.4	Safety Requirements Development	3					
7.5	HSI Requirements Development	4					
7.6	Requirements-Functions Traceability Update	3					
7.7	A-Spec Cleanup & Submittal	2					
8	Risk Management Report	15	16-Mar	30-Mar			
8.1	Risk Definitions Development	2					
8.2	Risk Categorization	2					
8.3	Individual Risk Assessments	3					
8.4	Mitigation Steps Documentation	3					
8.5	Risk Progression Analysis	3					
8.6	Risk Report Cleanup & Submittal	2					
9	Draft Final Report	16	30-Mar	6-Apr			
9.1	Executive Summary Development	3					
9.2	Report Sections Integration	4					
9.3	Schedule Assessment & EVM Analysis	3					
9.4	Lessons Learned Documentation	2					
9.5	Next Steps Development	2					
9.6	Draft Final Report Cleanup & Submittal	2					
10	Final Report & Oral Presentation	11	6-Apr	25-Apr			
10.1	Final Report Revisions	2					
10.2	Presentation Development	3					
10.3	Presentation Materials Preparation	2					
10.4	Final Report & Presentation Submittal	2					
10.5	Oral Presentation Delivery	2					
		292					

Figure 1: Work Breakdown Structure Update

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EVM

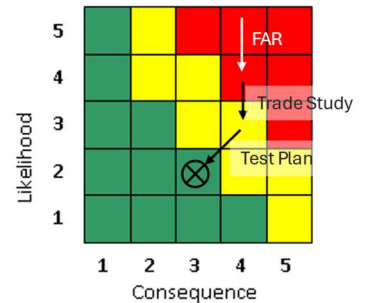
Table 3: EVM Update

Week	End of Week Date	BCWS	BCWP	ACWP	EVM Analysis					
					Cost Variance	Schedule Variance	SPI	CPI	TCPI (BAC)	EAC
1	1/26/2025	8	50	5	45	42	6.25	10.00	0.84	29.2
2	2/2/2025	26	50	30	20	24	1.92	1.67	0.92	175.2
3	2/9/2025	44	83.75	55	28.75	39.75	1.90	1.52	0.88	191.8
4	2/16/2025	62	83.75	65	18.75	21.75	1.35	1.29	0.92	226.6
5	2/23/2025	80	111.5	100	11.5	31.5	1.39	1.12	0.94	261.9
6	3/2/2025	98	111.5	120	-8.5	13.5	1.14	0.93	1.05	314.3
7	3/9/2025	116	165.25	150	15.25	49.25	1.42	1.10	0.89	265.1
8	3/16/2025	134	185.5	177	8.5	51.5	1.38	1.05	0.93	278.6
9	3/23/2025	152	174.5	162	12.5	22.5	1.15	1.08	0.90	271.1
10	3/30/2025	170	185.5	182	3.5	15.5	1.09	1.02	0.97	286.5

Appendix B – Risk Updates

Risk 1 – Navigation System Failure

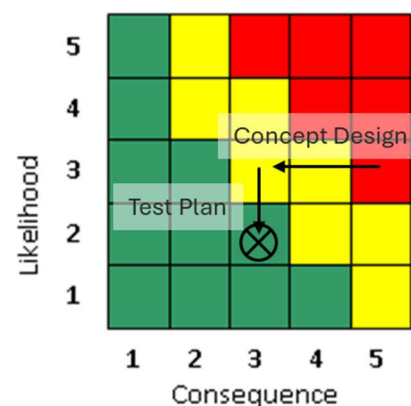
Risk Title	Navigation System Failure	
Description:	IF the GPS integration fails to meet accuracy requirements THEN autonomous positioning during refueling will be compromised, leading to unsafe operating conditions.	
Initial Assessment:	Likelihood:	5
	Consequences:	4
	Description of Consequences if realized	- Autonomous positioning failures would make it difficult to establish rough positioning of the tanker aircraft. Putting the mission, crew members, and others aboard at risk



Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	5	4	Initial assessment
2	RAR	Impose requirements for implementation of backup navigation system. See: ANGARS-22, Inertial Navigation.	5	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	FAR	Defined F6: Operate in Adverse Conditions behavior	4	4	Likelihood decreases due to implementation of functionality to switch sensors during conditions assessed as adversarial. Further definition of F6.4 should be done to burn down likelihood further.
4	Trade Study	Conducted trade on Navigation System to identify and choose navigation solution which would reduce the likelihood of navigation system failure	3	4	Likelihood reduces significantly. Solution chosen provides mitigation, reliability, redundancy, and accuracy to existing navigation system. Solution does not modify the consequence of risk should the accuracy risk be realized.
5	Test Plan	Implementation of phased integration testing approach with dedicated navigation system lab environment prior to aircraft installation	2	3	Both likelihood and consequence decrease. Early identification of navigation issues becomes possible through lab testing. The phased approach allows for resolving conflicts before aircraft installation, reducing both the likelihood of failures and the potential impact on overall project timelines and costs. The dedicated environment enables thorough testing of navigation components in GPS-denied conditions, validating redundant systems and sensor switching capabilities before flight testing.

Risk 2 – Communications System Failure

Risk Title	Communication System Failure	
Description:	IF secure data link is lost during operation THEN operational delays or mission failure may occur, potentially requiring manual intervention.	
Initial Assessment:	Likelihood:	3
	Consequences:	5



Risk 2: Communication System Failure

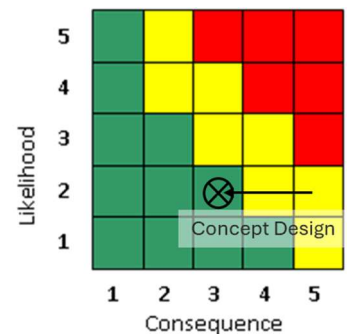
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	Description of Consequences if realized	- Comms going down would result in mission failures. Potential critical situational awareness updates may not be conveyed between ANGARS to ground or receiving aircraft, or vice-versa.
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Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	3	5	Initial assessment
2	RAR	Impose requirements defining design-level communication up-time constraints. See: ANGARS-33, Communication Uptime.	3	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	Concept Design	Imposed requirements to define redundant communication channels/modes for SATCOM and nominal comms. Implemented solution into design space through concept design	2	3	Likelihood that primary comms goes down has not been impacted due to the fact that primary comms is still susceptible to attack from adversaries. Consequence decreases significantly due to the fact that we now have a backup mechanism in the event of an attack on our primary comms mode, we have a secondary antenna which operates via different characteristics.
4	Test Plan	Implementation of phased integration testing approach with dedicated communications lab environment prior to aircraft installation	2	2	Consequence decreases. Early identification of communication issues becomes possible through lab testing. The phased approach allows for resolving conflicts before aircraft installation, reducing both the likelihood of failures and the potential impact on overall project timelines and costs. The test plan enables validation of redundant communication channels in simulated adverse conditions, confirming proper failover mechanisms function as designed.

Risk 3 – Collision Avoidance System Failure

Risk Title	Collision Avoidance System Failure	
Description:	IF autonomous collision avoidance fails to activate when needed THEN there is an increased risk of mid-air collisions.	
Initial Assessment:	Likelihood:	2
	Consequences:	5
	Description of Consequences if realized	- Autonomous collision avoidance system failures could result in catastrophic consequences such as bird strikes and fuel boom strikes.



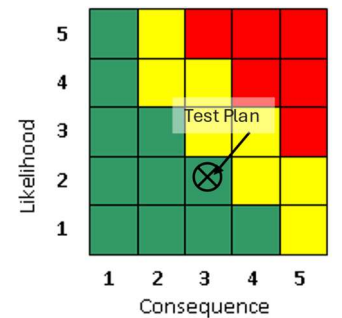
Risk 3: Collision Avoidance Failure

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	2	5	Initial assessment

2	RAR	Impose requirements defining design-level collision expectations. See: ANGARS-53, Collision Expectancy	2	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	Concept Design	Implementation has been designed and imposed to implement a solution of triple-redundant flight control computers and dual-redundant flight processors	2	3	Consequence reduces significantly due to the implementation of redundant collision avoidance systems and sensors.
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Risk 4 – Integration Challenges

Risk Title	Integration Challenges	
Description:	IF there are delays in integration with KC-46/NGAS platforms THEN schedule overruns and increased costs will impact project delivery	
Initial Assessment:	Likelihood:	3
	Consequences:	4
	Description of Consequences if realized	- Integration issues could result in significant schedule and cost risks, including overrun/budget. - Potential implications to existing ANGARS architecture or modifications to legacy hardware.



Risk 4: Integration Challenges

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	3	4	Initial assessment
2	RAR	Impose requirements defining design-level integration expectations. See: ANGARS-125, KC-46 Integration	3	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	Test Plan	Implementation of phased integration testing approach with dedicated integration lab environment prior to aircraft installation	2	3	Likelihood decreases significantly as early identification of integration issues becomes possible through lab testing. Phased approach allows for resolving conflicts before aircraft installation, reducing schedule risk. Consequence also decreases as potential integration problems can be addressed earlier in the development cycle, minimizing the impact on overall project timelines and costs.
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Appendix C – References

ICAO standard for atmospheric effects: <https://aiac.ma/wp-content/uploads/2018/01/Manuel-de-l%E2%80%99atmosph%C3%A8re-Type-OACI-Doc7488-1.pdf>

MIL-STD-810 temperature testing ranges/stds: <https://www.trentonsystems.com/en-us/resource-hub/blog/mil-std-810-temperature-testing>

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MIL-STD-882E, systems safety reference:

https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36027

Operational availability references: <https://www.osti.gov/servlets/purl/1315000>

Human system interfacing ref: <https://www.iso.org/standard/57992.html>